On the Case for Better Measures of Human Flourishing in the Age of AI

By Daniel T. Ketterer

1. What we measure shapes what we build for

I want to make a general argument, not a partisan one. The argument is this. The numbers a society commits to publishing every quarter become the numbers its institutions optimize. The metric is not a thermometer placed innocently against the room. It is a thermostat. It tells the furnace when to fire and when to rest. To measure something at scale, with statutory regularity, is already to govern it.

This is the central message of the 2009 Stiglitz-Sen-Fitoussi Commission, and it is restated explicitly in their 2018 follow-on report for the OECD: "what we measure affects what we do." Theodore Porter, in 'Trust in Numbers', traces the rise of quantification not as a downstream byproduct of scientific success, but as a "technology of distance" adopted where elite judgment is suspect and accountability must be made legible to strangers. Jerry Muller, in 'The Tyranny of Metrics', documents how this legibility, once institutionalized, displaces the very judgment it was meant to discipline. Charles Goodhart's law, "any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes," and Donald Campbell's law, that quantitative indicators distort the processes they monitor in proportion to their use, are not just warnings about gaming. They are warnings about ontology. The measure becomes the world the institution can see.

The rigorous version of "metrics are not neutral" is therefore not a complaint that GDP fails to capture beauty or love. It is a structural claim about feedback. Whatever metric a policy regime publishes and rewards, that is the variable along which the regime will tune itself. Everything orthogonal to that variable becomes invisible to the controller, and over time, invisible to the controlled.

That is the central premise. If you grant it, the rest follows.

2. Why current proxies are inadequate

GDP was never designed to measure welfare. Simon Kuznets, who built the national income accounts for the US Department of Commerce in the 1930s, warned in his 1934 report to Congress that "the welfare of a nation can scarcely be inferred from a measurement of national income." The warning was set aside because GDP was useful for what it was designed for, which was mobilization for the Second World War and the management of postwar industrial output. We have been governing wellbeing with a wartime production index for eighty years.

The standard alternatives sit on a spectrum. The Human Development Index, introduced by UNDP in 1990, adds health and education to income but remains a coarse three-variable composite. The OECD's Better Life Index publishes eleven domains but has no agreed aggregation. The World Happiness Report's Cantril Ladder gives a single self-reported life-evaluation digit zero through ten and is cheap to administer, but the question is hedonic and global. Bhutan's Gross National Happiness uses thirty-three indicators across nine domains, weighted by a method recognizable from Alkire and Foster's multidimensional poverty work. The Happier Lives Institute's WELLBY, one Well-Being-adjusted Life Year, allows direct comparison of interventions in a single subjective-wellbeing currency and is now the most actionable cost-effectiveness metric in global health philanthropy. Each of these is a real advance on GDP. None is sufficient on its own.

The empirical case for inadequacy has hardened in the last few years. Killingsworth, Kahneman and Mellers' 2023 adversarial collaboration resolved the apparent contradiction between Kahneman-Deaton's $75,000 plateau and Killingsworth's continued linear-log rise: the plateau holds, but only for the unhappy 20 percent of the population, and the underlying happiness distribution changes shape with income in ways neither original study modeled. This is decisive against the simple claim that money buys happiness, and equally decisive against the simple claim that it does not. Easterlin's paradox, that within rich countries long-run happiness does not track long-run GDP growth, has been re-examined by Stevenson and Wolfers and most recently by Oparina, Clark and Layard in their November 2024 CEP discussion paper 'The Easterlin paradox at 50'; the cross-section is positive, the time series in high-income countries is roughly flat. Both can be true at once because the construct is poorly resolved.

The deeper problem is structural, not parametric. Income, employment and hours are level variables of one quantity. Flourishing, on every theoretical account I know of, is a structure across many quantities. Ryff's six-dimensional model of psychological wellbeing, Keyes's social wellbeing, Diener's life satisfaction and Watson's positive affect, Sen's capability approach, the eudaimonic tradition from Aristotle through self-determination theory, all converge on the claim that what we are trying to track is the simultaneous functioning of several distinct life domains and the connections among them. A measurement framework that collapses this to a scalar will, by construction, miss the failures that matter most.

3. Why the gap matters more now, and is widening fast

The case for measurement reform was once an argument among economists about welfare theory. AI has made it urgent.

The empirical picture as of late 2025 is the following. The IMF's January 2024 staff discussion note 'Gen-AI: Artificial Intelligence and the Future of Work', authored by Cazzaniga, Jaumotte and colleagues, estimates that "almost 40 percent of global employment is exposed to AI," rising to 60 percent in advanced economies because of the prevalence of cognitive-task-oriented jobs. Roughly half of those exposed jobs are projected to benefit from complementarity; the other half face substitution. Goldman Sachs' Joseph Briggs and Devesh Kodnani, in their March 26, 2023 Global Economics Analyst report 'The Potentially Large Effects of Artificial Intelligence on Economic Growth', estimate that generative AI could "expose the equivalent of 300mn full-time jobs to automation" globally, with about 7 percent of US workers fully substituted and 63 percent complemented in their headline scenario. The 300 million figure is an exposure estimate, not a prediction of net job loss; the authors are explicit that historical reinstatement has offset earlier waves.

Acemoglu's 'The Simple Macroeconomics of AI', published in Economic Policy in January 2025, is more conservative on aggregate productivity, projecting about a one-percentage-point increase in US TFP over a decade, but his task-based framework with Restrepo is unambiguous about the distributional mechanism: automation expands the set of tasks performed by capital, and unless new-task creation, the "reinstatement effect," keeps pace, the labor share falls and wages of displaced groups drop. Brynjolfsson, Chandar and Chen's August 2025 Stanford Digital Economy Lab paper 'Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence', using ADP payroll data from a firm that "provides payroll services for firms employing over 25 million workers in the US," documents that "early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks," concentrated in roles where AI automates rather than augments tasks. Anthropic's Economic Index, in its fourth report published January 2026 and covering Claude usage data through November 2025, finds that Claude is now used for at least a quarter of tasks in 49 percent of jobs sampled, that augmentation has slightly overtaken automation in consumer-facing usage but automation dominates in API traffic, and that the tasks AI handles on average require 14.4 years of education versus an economy-wide average of 13.2, a "deskilling" pattern in which AI selectively absorbs the higher-skill components of jobs. The fifth report, published March 2026 and covering data through February 2026, finds these patterns deepening, with API automation growing fastest in business sales and customer service workflows.

I do not want to overstate this. The aggregate labor market has not collapsed. Through 2025, total employment in most advanced economies continues to grow. Humlum and Vestergaard's 'Large Language Models, Small Labor Market Effects' (BFI Working Paper 2025-56 / NBER w33777, April 2025), covering 11 exposed occupations across 25,000 workers and 7,000 workplaces in Denmark, "estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after the launch of ChatGPT." What is happening is sharper and more specific than the headlines: a compositional shift concentrated at entry level, in cognitive-task-heavy occupations, with the early-career rungs of established career ladders being removed faster than new rungs are being built. Predictions further out are uncertain and contested. I flag those as predictions, not facts.

But the policy-feedback argument from Section 1 does not require certainty about the trajectory. It requires only the observation that, if the trajectory is anything like what these studies suggest, the standard labor-market indicators will systematically understate what is happening to people. An unemployment rate of 4 percent is compatible with a generation of twenty-five-year-olds whose careers never compounded. Median wage growth is compatible with the hollowing-out of meaningful work into validation and oversight. A flat labor-force-participation rate is compatible with a population that has become structurally surplus to economic production and learned to stop reporting that they are looking.

The instruments that institutions stare at every month are about to become less informative just as the underlying changes accelerate. That is the gap.

4. The specific failure mode: managed decline, measured by absence of complaint

If we do not change what we measure, I think the default policy response to AI-driven labor displacement is reasonably predictable. It will be income transfers, sized to keep displaced workers above some politically tolerable threshold, with success measured by whether the transfers preserve consumption and whether recipients report acceptable levels of distress on instruments that were designed to detect destitution rather than the absence of flourishing. Managed decline, measured by absence of complaint.

I do not say this is what advocates of basic income intend. Many of them, including Banerjee and Duflo on the development side, are explicit that cash is necessary but not sufficient. I am describing the default of the institutional system, the path of least resistance under metrics that already exist.

The evidence on whether income transfers alone produce flourishing is more nuanced than either side of the UBI debate usually allows. The largest US study, the OpenResearch Unconditional Cash Study, gave 1,000 low-income adults aged 21 to 40 in Illinois and Texas $1,000 per month for three years against a control group of 2,000 who received $50 per month. Three NBER working papers in mid-2024 report the findings. Vivalt, Rhodes, Bartik, Broockman, Krause and Miller, in NBER w32719 'The Employment Effects of a

Guaranteed Income', find that "the transfer caused total individual income excluding the transfers to fall by about $1,800/year relative to the control group and a 4.1 percentage point decrease in labor market participation. Participants reduced their work hours as a result of the transfers by 1-2 hours/week... Despite asking detailed questions about amenities, we find no impact on quality of employment, and our confidence intervals can rule out even small improvements." Miller and colleagues, in NBER w32711 'Does Income Affect Health?', find that "the cash transfer resulted in large but short-lived improvements in stress and food security," and over the three-year horizon, "we find no effect of the transfer across several measures of physical health, and we can rule out even very small improvements... the transfer did not improve mental health after the first year and by year 2 we can again reject very small improvements." On agency, the OpenResearch summary reports recipients were 14 percent more likely to pursue education or training and showed increased entrepreneurial interest, but these aspirations did not translate to measurable gains in degree attainment or business ownership.

These are not nothing. Cash buys real things, and the agency findings are genuine. But the headline pattern is that transfers improved consumption, modestly reduced labor supply without finding offsetting gains in job quality or human capital, and produced wellbeing benefits that faded by year two. The authors' own conclusion is restrained: "more targeted interventions may be more effective at reducing health inequality... at least for the population and time frame that we study."

Set this against what we know about the psychological structure of unemployment. Marie Jahoda's latent-deprivation theory, formulated in the 1930s on the Marienthal study and revised in 1982, claims that paid work supplies not only income but five latent functions: time structure, social contact, status, collective purpose, and forced activity. Paul, Scholl, Moser, Zechmann and Batinic's 2023 meta-analysis in Frontiers in Psychology, the first published meta-analysis of the latent-deprivation model, finds substantial support across decades of empirical work: the latent functions do mediate a sizable share of the unemployment-mental-health relationship, with status and time structure the strongest contributors. Clark, Diener, Georgellis and Lucas' fifteen-year German panel analysis, published in Psychological Science in 2004 and extended in the 2008 Economic Journal paper, finds that adaptation to unemployment is, for men, essentially absent: life satisfaction does not return to baseline even after reemployment. This is one of the most replicated scars in subjective-wellbeing research. Case and Deaton's 'Deaths of Despair and the Future of Capitalism', controversial in detail but striking in pattern, links mortality from suicide, drug overdose, and alcoholic liver disease in the US since the late 1990s to the destruction of working-class meaning structures alongside wage stagnation.

The empirical claim, then, is that work supplies something cash does not. The normative claim follows. A policy that replaces wages with transfers and measures success by income preserved is optimizing along one axis of a multidimensional thing and ignoring the others. It will, with very high probability, deliver something the official numbers register as a success while delivering something people, in their actual lives, experience as a slow erasure.

I am not arguing against transfers. I am arguing against measuring them with the instruments we have.

5. What a better framework would have to do

Stipulate that the goal is to capture flourishing, in the Aristotelian-eudaimonic sense, at population scale, in a way useful for policy. Then the framework has to satisfy at least five constraints simultaneously, and the existing alternatives each satisfy a subset.

It has to be multidimensional rather than scalar. The positive manifold across wellbeing instruments, the empirical finding that Ryff, Keyes, Diener, Watson and related scales correlate strongly but not perfectly, is consistent with both a single underlying latent factor and a coupled-domain dynamical system; the policy question of where deprivation is concentrated requires that the framework can decompose. Alkire and Foster's dual-cutoff multidimensional methodology, used in the global Multidimensional Poverty Index, shows how to do this rigorously for poverty; the technique generalizes.

It has to be structural, sensitive to coupling and configuration, not just levels. Two populations with the same average on a composite can have utterly different distributions of within-person deprivation. Compensatory scoring of a general factor, where high social wellbeing offsets low purpose, can conceal capability failures at policy-relevant rates. My own work in MIDUS 2, applying Alkire-Foster thresholds to seventeen wellbeing indicators across eight instrument families, finds that 58.5 percent of respondents fall below threshold on at least one domain even when the overall composite looks healthy. That number tells you something the composite cannot.

It has to be eudaimonic, not just hedonic. Ryff's six dimensions, Ryan and Deci's self-determination theory, Steger's meaning-in-life work, and the Global Flourishing Study's six-domain framework all converge on the claim that flourishing involves purpose, autonomy, environmental mastery, and relationships, not just affect. Killingsworth-Kahneman shows that even hedonic measures are heterogeneous below the surface. A framework that ignores eudaimonic content will, in an AI-displacement scenario, miss exactly what is being lost.

It has to be stable enough for policy use and responsive enough to capture real change. The OECD's 2025 guidelines update on subjective wellbeing measurement, building on a decade of cross-cultural validity work, gives a defensible practical baseline. The Happier Lives Institute's WELLBY durability research, examining how long psychotherapy-induced gains persist relative to cash-transfer-induced gains, gives a model for how to integrate dynamics.

It has to be identifiable. By that I mean the construct must be testable in causal-inference frameworks: cross-sectional covariance does not determine the drift matrix of a continuous-time linear system, which is why longitudinal panels like the Global Flourishing Study, with 200,000 respondents across 22 countries and five planned waves through 2028, are the right substrate. Without identification, every measurement framework collapses to a politically negotiated index.

The general factor of flourishing I have been developing, which I call f, formalized as systemic coupling strength across life domains and tested via bifactor confirmatory factor analysis with auxiliary dynamical-systems modeling, is one candidate construct that tries to satisfy these constraints. So are the Global Flourishing Study's multidimensional framework, OECD's expanding subjective-wellbeing apparatus, and HLI's WELLBY-plusstructure agenda. I do not claim f is the answer. I claim the criteria are the right criteria, and the field is converging on them.

6. Engaging the strongest objections

The first objection: subjective measures are too noisy or culturally relative for policy. The strongest scholarly version of this is Robert Cummins' work in 'Measuring and Interpreting Subjective Wellbeing in Different Cultural Contexts', which documents pervasive response-bias differences across cultures. The response is empirical, not rhetorical. The OECD's 2025 report on globally inclusive measures of subjective wellbeing reviews the cross-cultural-validity literature and concludes that "by and large" the evidence supports rather than undermines the validity of the OECD core measures, though it flags genuine open issues. The Global Flourishing Study has demonstrated measurement invariance across 22 countries on its core flourishing items. Noise is real; comparison is hard. But "harder than GDP" is not a knockdown argument against using a measure. GDP itself, properly examined, has comparability problems across countries that we have learned to live with because we built institutions around handling them. We can do this for flourishing.

The second objection: GDP correlates well enough with most things that matter that the marginal value of new measures is small. The strongest version of this is the Stevenson-Wolfers reading of the cross-country evidence, on which income predicts life evaluation strongly. The response is that the correlation is being asked to do a job it cannot do once the link between aggregate output and individual wellbeing is being mediated less and less by labor income. If two-thirds of US occupations are exposed to AI substitution in the next decade in any meaningful fraction, GDP can keep rising while the labor-share-of-flourishing channel collapses. The marginal value of new measures is not small in that scenario. It is the only thing standing between policy and the wrong dashboard.

The third objection: measurement reform is a distraction from material redistribution. I have heard this from people I take very seriously on the left. The response is that the order is wrong. Reform of measurement is not a substitute for redistribution; it is the necessary precondition for redistribution that is responsive to anything beyond consumption preservation. The Marienthal observation, that workers who keep their wages and lose their work decline in measurable ways anyway, is the empirical fact that should govern this debate. Transfers without a measurement framework that tracks meaning, agency, structure, and connection are not the absence of policy. They are a policy that will be optimized for whatever cheaper proxies the institutional system can stomach.

The fourth objection: Goodhart's law applies even more strongly to flourishing measures, because they are softer, more game-able, more open to political capture than GDP. This is the most serious objection. The honest answer is that any flourishing metric used for policy will be gamed in ways we cannot now anticipate, that this is a feature of all metrics used at scale, and that the appropriate response is not to refuse to measure but to design constructs that are structural rather than reductive, hard to game without producing the underlying improvements, and embedded in pluralistic dashboards rather than singular targets. Manheim and Garrabrant's taxonomy of Goodhart variants is the right place to start engineering robustness. Bifactor models in particular have a useful property: gaming requires moving an entire correlated set of indicators in concert, which is closer to actually improving flourishing than gaming a single dimension.

7. Conclusion

What we measure shapes what we build for. The metrics we have were built for an industrial economy where wages bought lives and employment supplied structure. That economy is unwinding in measurable ways, fast, in entry-level cognitive work first. If we keep measuring as we do, the institutional response will be to preserve consumption, ratify the absence of complaint, and call it a soft landing. The official numbers will glow green. The lives underneath will hollow out.

The argument for better measurement is not aesthetic. It is structural and urgent. We need instruments that are multidimensional, sensitive to coupling and configuration, eudaimonic, identifiable, and stable enough to govern with. We have several candidates and a growing empirical infrastructure, the Global Flourishing Study, the WELLBY, the Alkire-Foster apparatus, and the OECD framework, to build on. The question is whether we adopt them before or after the wrong dashboard becomes the only one that anyone reads.

I would rather we did it before. The work is doable. The cost of getting it wrong is a generation of people for whom the official record will say everything is fine.